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* 1. Convert the data file SVEDKA.csv to a SAS dataset. Study the data along with the data dictionary and write a summary about it.;  
libname assignmt1 '\\apporto.com\dfs\IIT\Users\gauravsinha_iit\Desktop\Quantitative Marketing Models';
```

```
proc import out= assignmt1.svedkadata  
    datafile= "\\apporto.com\dfs\IIT\Users\gauravsinha_iit\Desktop\Quantitative Marketing Models\Assignment 1\svedka.csv"  
    dms= csv;  
getnames=yes;  
datarow=2;  
run;  
  
/* looking at the content of the dataset imported */;
```

```
proc contents data= assignmt1.svedkadata;  
run;  
  
/* printing some of the records of the dataset */;
```

```
proc print data=assignmt1.svedkadata (obs=10);  
run;
```

```
* 2. Generate summary statistics for the data. What can you infer about the data at this stage?;
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```
proc means data = assignmt1.svedkadata;  
var TotalSales Domestic DollarSales PricePerUnit Mag News Outdoor Broad Print Tier TotalMinusSales Firstintro Marketshare;  
run;
```

```
proc freq data = assignmt1.svedkadata;  
table brandname tier;  
run;
```

```
proc corr data = assignmt1.svedkadata;  
var totalsales priceperunit print domestic outdoor broad firstintro marketshare totalminusales tier;  
run;
```

```
*3. Run a linear regression of sales on prices. Write out the regression equation. What can you conclude about Vodka prices?;  
/* Model 1 */
```

```
proc reg data = assignmt1.svedkadata plots=none;  
model totalsales = priceperunit;  
run; quit;
```

```
* 4. Run a regression which includes price and advertising. What variables in the data represent advertising?
Describe your methodology for choosing your model specification. Write out the regression equations for your specification.
Comment on your results. ;
/* Model 2 */
proc reg data = assignmt1.svedkadata plots = none;
model totalsales = priceperunit broad outdoor mag news;
run; quit;
```

```
* 5. The dataset classifies the brands into three quality tiers.
Tier 1 brands: Chopin, Belvedere, Grey Goose, Level, Ketel One, Absolut, Stolichnaya, Pravda, and Ciroc.
Tier 2 brands: Smirnoff, Sky, Finlandia, Three Olives, Fris, Seagrams, Polar Ice, Van Gogh, Tanqueray, Pearl, Iceburg, and Svedka
Tier 3 brands: All remaining brands
```

Create dummy variables for the Tier variable. How many dummy variables will you need?;

```
data assignmt1.svedkadata_updated;
set assignmt1.svedkadata;
dummy_tier1 = (Tier = 1);
dummy_tier2 = (Tier = 2);
run;

proc print data = assignmt1.svedkadata_updated (obs=10);
run;
```

```
* 6. Run a regression model to understand the influence of vodka quality tiers on sales by including Tier dummies to the above model.
What do the results tell you about Tier 1 and Tier 2 brands? Should Svedka management consider competing in Tier 1?;
```

```
/* Model 3 */
proc reg data = assignmt1.svedkadata_updated plots = none;
model totalsales = priceperunit broad outdoor mag news dummy_tier1 dummy_tier2;
run; quit;
```